

Social Networks: Assignment #3

Due on Thursday, January 4, 2026

Dr. Masoud Asadpour 15:00am

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(a)	Using the standard graph reduction algorithm based on positive connected components, construct the reduced graph G' . Clearly specify the vertices of G' (i.e., the supernodes), listing the original nodes contained in each supernode.	4

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Question 1

Graph Reduction in Large and Irregular Signed Networks

(a) Using the standard graph reduction algorithm based on positive connected components, construct the reduced graph G' . Clearly specify the vertices of G' (i.e., the supernodes), listing the original nodes contained in each supernode.

After running the graph reduction algorithm we reach 5 super-nodes listed below:

- super-node A containing nodes {6, 7, 8}
- super-node B containing nodes {1, 2, 3, 4, 5}
- super-node C containing nodes {14, 15, 16, 17, 18}
- super-node D containing nodes {10, 11, 12, 13}
- super-node E containing node {9}

References

- [1] ChatGPT, *Chat session used for assignment guidance*, <https://chatgpt.com/share/69518d24-9800-800d-8fe8-185a102b5f10>, accessed December 28, 2025.